



Integrated bioinformatic evaluation unveils a cellular senescence gene signature as a poor prognostic factor in hepatocellular carcinoma

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Abstract

Cellular senescence (CS) plays a crucial role in various diseases, but its role in hepatocellular carcinoma (HCC) remains unclear. CS-related genes were clustered to identify subtypes. A risk score was constructed and validated in three independent cohorts. Associations with clinical characteristics, tumor immune microenvironment, mutation status, heterogeneity, and treatment efficacy were analyzed. Single-cell analysis was used to examine risk score distribution, and machine learning algorithms along with a nomogram were applied to assess prognostic value. Three CS subtypes were identified, with subtype 1 showing the worst prognosis. High risk score was associated with advanced clinical stage and grade, poor prognosis, and an immunosuppressive microenvironment driven by regulatory T cells. It also correlated with higher tumor mutations (notably TP53) and increased heterogeneity. High-risk patients showed poor response to sorafenib and transcatheter arterial chemoembolization (TACE) and may benefit less from immunotherapy. At single-cell level, the risk score was predominantly expressed in malignant hepatocytes and linked to cell stemness. The CS-related risk score is a potential prognostic indicator for poor outcomes in HCC, playing a significant role in tumor progression and offering potential value for clinical diagnosis and prediction of treatment response.

Keywords Cellular senescence · Hepatocellular carcinoma · Tumor immune microenvironment · Immunotherapy

Abbreviations

CS	Cellular senescence
HCC	Hepatocellular carcinoma
RCC	Renal cell carcinoma
STAD	Stomach adenocarcinoma
LGG	Brain Lower Grade Glioma
HAGR	Human Ageing Genomic Resources
TCGA	The Cancer Genome Atlas
GEO	Gene Expression Omnibus
ICGC	International Cancer Genome Consortium
TIGER	Tumor Immunotherapy Gene Expression Resource
CROST	Comprehensive Repository of Spatial Transcriptomics
CPTAC	Clinical Proteomic Tumor Analysis Consortium
TACE	Transcatheter arterial chemoembolization
WGCNA	Weighted gene co-expression network analysis
GO	Gene ontology
KEGG	Kyoto encyclopedia of gene and genome
LASSO	Least absolute shrinkage and selection operator

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MATH	Mutant-allele tumor heterogeneity
Fegs	Functional gene expression profiles
PCA	Principal component analysis
ssGSEA	single sample gene set enrichment analysis

Introduction

Hepatocellular carcinoma (HCC) is the most common type of liver cancer and ranks as the third leading cause of cancer-related mortality worldwide, with a five-year survival rate of less than 20%. The incidence of HCC has been steadily increasing in recent years, and it is projected that by 2040, there will be approximately 1.4 million new cases and 1.3 million deaths globally [1, 2]. The treatment for early-stage HCC patients involves surgery combined with radio-chemotherapy, but most patients have reached the advanced stage by the time they are diagnosed, and their prognosis is significantly poor due to their susceptibility to metastasis, recurrence and limited treatment options [3, 4].

Cellular senescence (CS), characterized by an array of heterogeneous stress responses, manifests as a state of irreversible cell cycle arrest. It can be triggered by various factors, including telomere shortening, epigenetic perturbations, and the expression of specific oncogenes [5]. Senescent cells in the tumor microenvironment (TME) can promote tumor proliferation by inducing pathological processes such as epithelial-mesenchymal transition and tumor vascularization via the secretion of senescence-associated secretory phenotypes [6, 7]. During cellular senescence, it has been observed that the activation of the immune cell suppressor network, which encompasses regulatory T (Treg), regulatory B (Breg), and regulatory NK (NKreg) cells, can result in a disruption of the homeostasis of the immune microenvironment [8]. Colucci et al. [9] demonstrated that senescent tumor cells can promote cancer progression through the acquisition of stem cell-like properties and re-entry into the cell cycle. Nevertheless, technical limitations have led previous studies to focus on only one or two senescence genes, ignoring the fact that the antitumor effects of these genes are controlled by numerous tumor suppressors that act in a tightly coordinated manner. Furthermore, the accumulation of transcriptomic and genomic data provides an excellent resource for comprehensive analysis of senescence genes and their effects on immune regulation. Thus, we could greatly enhance our understanding of cancer immunity by identifying TME cell infiltration characteristics influenced by multiple senescence genes. Consequently, the exploration into the pertinent role of cellular senescence in the tumorigenicity and progression of HCC may establish a novel paradigm for tumor immunotherapy.

In our study, we identified three CS-related subtypes by consensus clustering and screened for the genes associated with poor prognostic subtypes. The CS risk score was constructed by LASSO and Cox regression and validated by three independent cohorts. In addition, we investigated the potential of the risk score to evaluate immune infiltration, tumor mutations, immunotherapy, and to predict clinical prognosis in HCC patients. The study may provide a robust insight for the diagnosis and prognosis of clinical HCC patients.

Methods

Data source

543 CS-related genes were acquired in the HAGR database [10]. Transcriptome data were obtained from The Cancer Genome Atlas (TCGA), Gene Expression Omnibus (GEO), Tumor Immunotherapy Gene Expression Resource (TIGER) [11], and International Cancer Genome Consortium (ICGC). TCGA-LIHC was downloaded as a training cohort, and ICGC-LIRI-JP, GSE14520, and GSE76427 cohorts were downloaded as validation cohorts. In addition, we explored the ability of the risk score in predicting patient response to treatment with sorafenib and Transcatheter Arterial Chemo-embolization (TACE) using TCGA-LIHC and GSE109211 for sorafenib, and GSE14520 and GSE104580 for TACE, as well as immunotherapy response using the GSE238264, E-MTAB-6270, GSE67501, and PRJEB25780 cohorts. All transcriptomic expression data were log2 transformed prior to downstream analysis.

Single-cell dataset from the GSE166635 cohort was processed by the following criteria: (1) retaining cells expressing genes in the range of 500 to 10,000; (2) removing genes expressed in less than 3 cells; (3) retaining cells with gene expression between 1000 and 100,000; (4) retaining cells with less than 15% of mitochondrial genes. A total of 23,725 genes and 17,853 cells were obtained for subsequent analysis.

The protein expression data of the genes were obtained from the PDC000198 dataset of the Clinical Proteomic Tumor Analysis Consortium (CPTAC) database. The following processes were performed: (1) deleting genes with deletion of more than 50%; (2) filling of missing values by the KNN algorithm of the “impute” package (k=10, default); (3) between-sample normalization using the normalizeBetweenArrays function of the ‘limma’ package with the “quantile” method.

Spatial transcriptomic data for the VISDS000519 sample were obtained from the CROST database [12]. The dataset had been pre-processed to remove background noise and

low-quality spots, ultimately retaining 19,951 genes and 4,654 spots for subsequent analysis.

Differential expression and consensus clustering of CS-related genes

Differential expression analysis of the 543 CS-associated genes was performed using the “limma” package. Linear models were fitted using the `lmFit` function, followed by empirical Bayes moderation with `eBayes`. Differentially expressed genes were identified with the criteria of $|\log FC| \geq 1$ and adjusted P-value < 0.05 . The differentially expressed genes are used for unsupervised consensus clustering through the “ConsensusClusterPlus” package [13]. Single sample gene set enrichment analysis (ssGSEA) was performed by the “GSVA” package to calculate the pathway enrichment activity of the TCGA cohort and to compare the differences in pathways among the three subtypes [14].

Screening for subtype-related CS genes

Weighted Gene Co-expression Network Analysis (WGCNA) was performed on differentially expressed senescence genes by the “WGCNA” package to screen for genes closely related to subtype 1 [15]. First, the best soft threshold β was set to 4, and a total of 10 non-gray modules were obtained. Then, the correlation between each module and the subtype was calculated, and the modules with a high correlation with subtype 1 were selected. The modules were enriched with Gene Ontology (GO) and Kyoto Encyclopedia of Gene and Genome (KEGG) by the “clusterProfiler” package for the hub genes contained in the modules [16].

Construction of CS-related risk score

The hub genes screened by WGCNA were further analyzed using Least Absolute Shrinkage and Selection Operator (LASSO) regression with 10-fold cross-validation via the “glmnet” package, and the optimal penalty parameter λ was determined as `lambda.min` [17]. Multivariate Cox regression analysis was then performed to identify genes with independent prognostic value. The identified genes were used to construct the CS-related risk score, and samples in each cohort were divided into high- and low-risk groups based on the median risk score. Time-dependent Receiver Operating Characteristic (ROC) curves and Decision Curve Analysis (DCA) curves were generated using the “timeROC” and “ggDCA” packages to evaluate the prognostic performance of the risk score across the four cohorts [18].

Immune-related analysis

Information on each immunomodulator was obtained through the TISIDB database to explore the relationship between risk score and each immunomodulator [19]. Correlation between risk score and immune cell infiltration explored by CIBERSORT algorithm [20]. In addition, We used seven algorithms, namely XCELL, TIMER, QUANTISEQ, MCPOUNTER, EPIC, CIBERSORT-ABS, and CIBERSORT, to calculate the relative proportions of immune cell infiltration in the TCGA-LIHC cohort. The “ImmuneSubtypeClassifier” package was used to calculate the immune subtypes for each of the TCGA cohort samples [21].

Mutation-related analysis

Analysis of mutation data from the TCGA cohort via the “maftools” package and presentation of the top 20 mutated genes for patients in the high and low-risk groups [22]. The Mutant-Allele Tumor Heterogeneity (MATH) scores of each sample were calculated through the “maftools” package to investigate the correlation between the risk score and MATH scores.

Single-cell analysis

Integration of GSE166635 cohort data via the “harmony” package to eliminate batch effects across samples [23]. Clustering of cells by the “seurat” package and dimensionality reduction of the data by the Uniform Manifold Approximation and Projection (UMAP) algorithm [24]. CytoTRACE algorithm is used to infer the stemness score of each cell, the higher the CytoTRACE score the stronger the cell stemness [25]. The “Scissor” package is used to infer the correlation between cells and phenotypes based on the survival status of patients in the TCGA-LIHC cohort and the phenotypic data of TP53 mutations, in which “scissor+” indicates a positive correlation with the inferred phenotype and “scissor-” indicates a negative correlation with the inferred phenotype [26].

Spatial transcriptomics analysis

“SpaCET” package [27] was employed to perform cell type deconvolution analysis on the VISDS000519 sample dataset, inferring the proportions of malignant cells, hepatocytes, macrophages, cancer-associated fibroblasts (CAFs), cholangiocytes, and endothelial cells in each spot, and their spatial distributions were visualized. Furthermore, the tumor-stroma interface region was identified and its spatial localization was demonstrated. Subsequently, the “Seurat”

package was utilized to read the sample data, and the SCTransform method was applied for data normalization. The risk score for each spot was calculated based on the following formula: $\text{Riskscore} = 0.0020983510 \times \text{NDRG1} - 0.0010218583 \times \text{PON1} + 0.0008621069 \times \text{SQSTM1}$. Then, the spatial distribution of the risk score was visualized, along with the spatial expression patterns of the target genes.

Oncology treatment

The “oncoPredict” package [28] was used for drug sensitivity prediction, with the GDSC2 dataset serving as the training set. Before the analysis, the IC50 values in the GDSC2 dataset were log-transformed. The TCGA-LIHC tumor expression data were preprocessed by removing low-expression genes and normal samples. Subsequently, the calcPhenotype function was applied for prediction. In addition, two GEO cohorts, GSE109211 and GSE104580, were used to explore the response of patients in different risk groups after treatment with sorafenib and TACE. To further validate the prognostic value of the risk score in these treatment populations, we analyzed survival outcomes in independent cohorts, including the TCGA-LIHC sorafenib-treated subgroup and the GSE14520 TACE-treated cohort. Moreover, to validate the predictive value of the CS risk score for immunotherapy response, we analyzed multiple independent external cohorts, including immunotherapy cohorts (E-MTAB-6270, GSE67501, PRJEB25780) and a combination therapy cohort (GSE238264). The detailed

information of all cohorts used for treatment response validation is summarized in Table S1.

Clinical prognostic characteristics

A nomogram integrating the risk score with clinicopathological features, including age, gender, tumor stage, and grade was constructed for the TCGA cohort using the “reg-plot” package. The accuracy of the nomogram model was validated using calibration curves (1-, 3-, and 5-year), time-dependent ROC curves, and decision curve analysis (DCA).

Statistical analysis

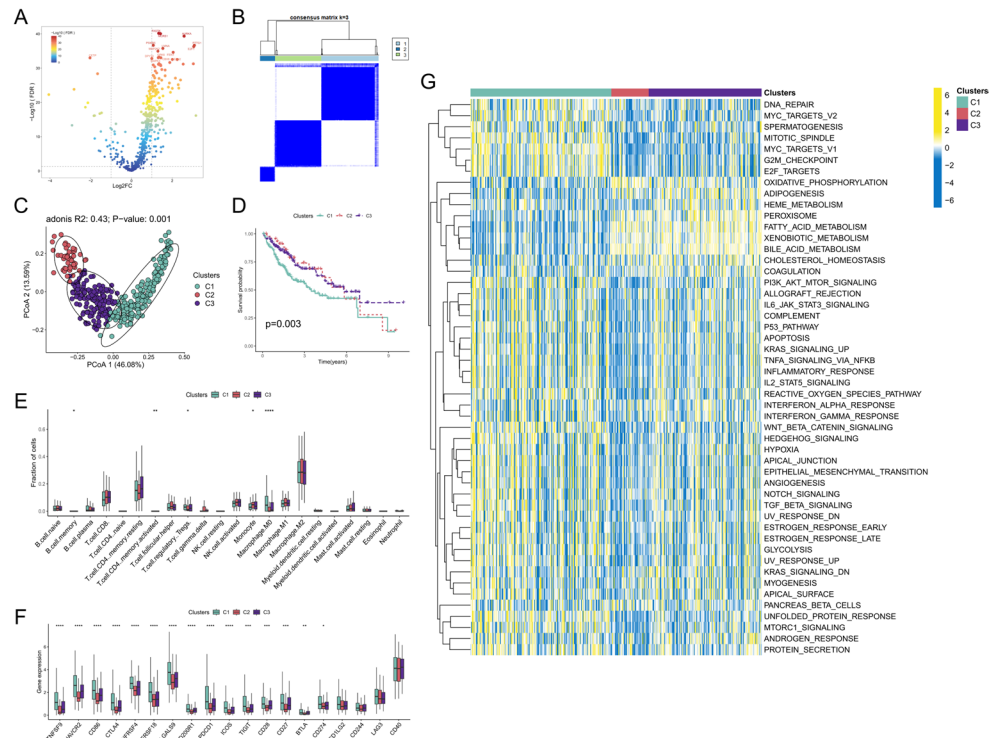
The study was analyzed by R software (v.4.2.2). Survival analysis by Kaplan-Meier (K-M) method, comparison between two groups by the Wilcoxon test, comparison between multiple groups by the Kruskal-Wallis test, correlation analysis by the spearman correlation test, significance defined as $P < 0.05$ (**** < 0.0001 ; *** < 0.001 ; ** < 0.01 ; * < 0.05).

Results

CS genes redefine the molecular typing of HCC

Differential analysis of 543 CS-related genes resulted in 117 up-regulated genes and 23 down-regulated genes (Fig. 1A).

Fig. 1 CS genes redefine the molecular typing of HCC. (A) Volcano plot of differential expression of CS-related genes. (B) Consensus clustering matrix with K=3. (C) Results of PCoA analysis based on 140 differential genes. (D) Kaplan-Meier survival analysis curves for the three subtypes. (E) Box plot of the proportion of 22 immune cell infiltrations for the three subtypes. (F) Box plots of the expression levels of immune checkpoint genes among the three subtypes. (G) Heatmap of pathway enrichment activity for the three subtypes



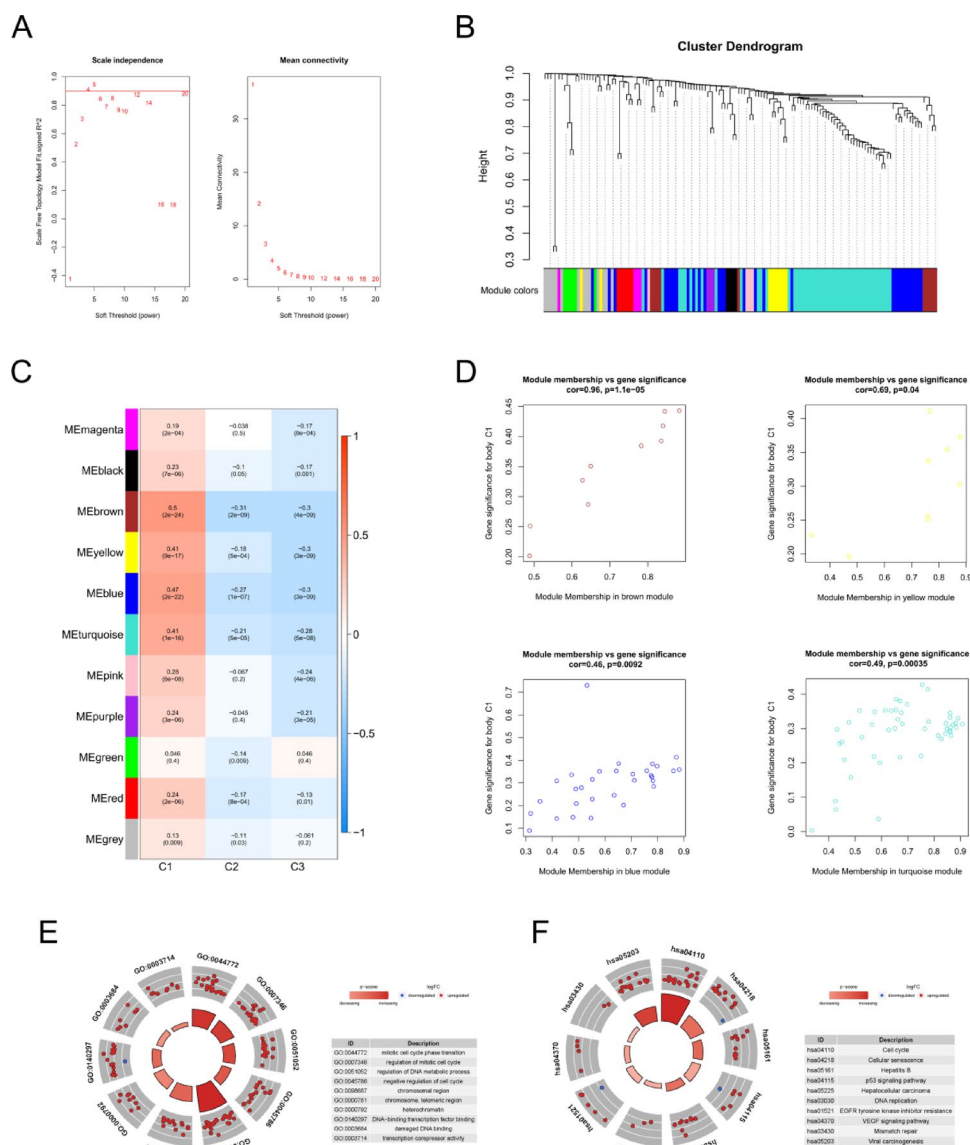
Unsupervised consensus clustering of patients based on 140 differentially expressed genes classified them into three subtypes C1, C2, and C3 (Fig. 1B, Figure S1A, B). The results of principal coordinate analysis (PCoA) and multivariate ANOVA showed that patients in the three subtypes could be effectively distinguished based on 140 differentially expressed genes (Fig. 1C, Figure S1C). Patients in the C1 subtype had a significantly poorer prognosis than those in the C2 and C3 subtypes ($p < 0.05$, Fig. 1D). Naturally, we proceeded to analyze the heterogeneity of the immune microenvironment of CS molecular subtypes. There was more regulatory T-cell and M0 macrophage infiltration in the C1 subtype, while monocyte infiltration was lowest in the C1 subtype (Fig. 1E). Consistently, immune checkpoint genes including TNFSF9, CTLA4, and PDCD1 were also highly expressed in the C1 subtype (Fig. 1F). Last, we analyzed the differences in signaling pathways of the three

molecular subtypes with GSVA, and MYC, DNA repair and cell cycle pathways were enriched in the C1 subtype (Fig. 1G). The above results indicate that the C1 subtype is the molecular subtype with the worst prognosis in HCC, and exists immunosuppressive microenvironment and cell proliferation-related pathways.

Molecular typing of CS in HCC has different gene expression patterns

To exploit the pivotal genes associated with subtype 1, WGCNA analysis was performed on 140 differential genes. To ensure the scale-free network structure $R^2 \geq 0.9$, the best soft threshold β was set to 4 (Fig. 2A), and a total of 10 non-gray modules were obtained (Fig. 2B). Among them, there were high correlations between brown, yellow, blue, and turquoise modules and subtype 1 (Fig. 2C). There was a

Fig. 2 Molecular typing of CS in HCC has different gene expression patterns. **(A)** Scale independence and average linkage for soft thresholds 1–20. **(B)** Cluster dendrogram constructed by weighted correlation coefficients and 10 non-gray modules. **(C)** Correlation heatmap of each module with subtypes. **(D)** Module membership (MM) of brown, yellow, blue and turquoise modules with gene significance (GS) of C1 correlation scatter plots. **(E)** GO functional analysis of subtype 1 related genes. **(F)** KEGG enrichment analysis of subtype 1 related genes



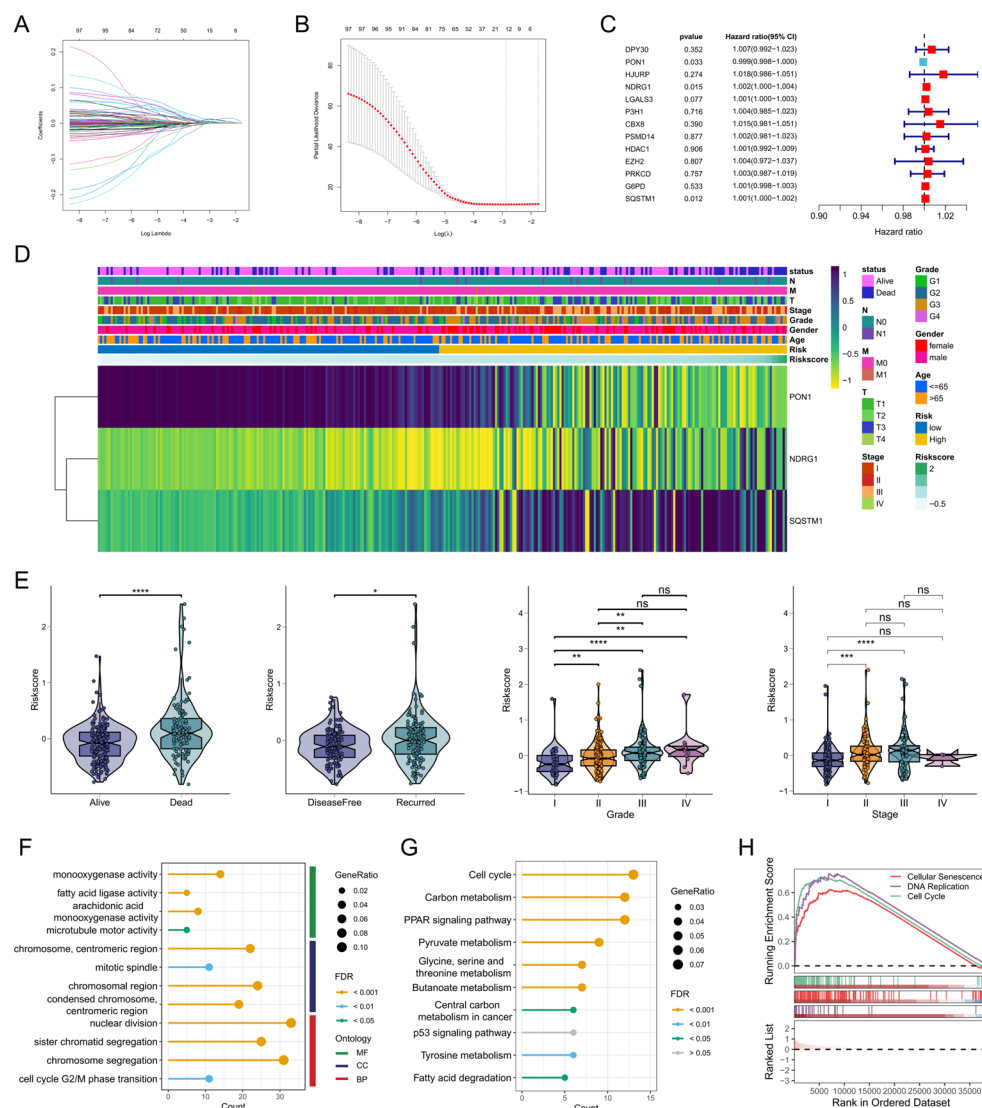
high correlation between the module membership (MM) of the four modules and the gene significance (GS) of the C1 subtype (Fig. 2D). GO function analysis of the four module hub genes showed that they were mainly associated with cell cycle mitosis, DNA metabolic process, and negative regulation of cell cycle (Fig. 2E), and the KEGG pathway was mainly enriched in cell cycle, cellular senescence, DNA replication, and other senescence-related pathways as well as cancer-related pathways such as hepatocellular carcinoma, p53 signaling pathway, VEGF and EGFR (Fig. 2F).

The CS risk score distinguishes the clinicopathological features of HCC and enforces different molecular functions

LASSO regression and multivariable Cox regression analysis of 99 pivotal genes obtained from WGCNA analysis revealed that three genes, PON1, NDRG1, and SQSTM1, had independent prognostic significance (Fig. 3A-C, $p < 0.05$). The

risk score was constructed by expression of the three genes with multifactorial COX regression coefficients (Riskscore = $0.0020983510 \times \text{NDRG1} - 0.0010218583 \times \text{PON1} + 0.0008621069 \times \text{SQSTM1}$), and patients were divided into high and low-risk groups according to the median value of the risk score. The expression of PON1 was lower in the high-risk group than in the low-risk group, and its high expression favored patient prognosis, whereas the expression of NDRG1 and SQSTM1 was higher than in the low-risk group, which is unfavorable to the prognosis of patients (Fig. 3D, Figure S1D-F). The risk score was significantly correlated with patient survival status, tumor recurrence, clinical grade, and clinical stage (Fig. 3E). In addition, GO and KEGG enrichment analysis revealed that differential genes in different risk groups were mainly enriched in cell cycle and metabolic-related pathways, which is consistent with known cellular senescence-related features [29, 30] (Fig. 3F, G). Similarly, GSEA analysis also revealed

Fig. 3 The CS risk score distinguishes the clinicopathological features of HCC and enforces different molecular functions. (A) LASSO regression coefficient of subtype-related genes. (B) Adjustment parameter (λ) calculated based on 10-fold cross-validation. (C) Multivariate COX regression forest plot of 13 genes screened by LASSO regression. (D) Expression heatmap of 3 genes in different risk groups. (E) Violin plot of the risk score in different clinical features. (F) GO enrichment analysis of differential genes in different risk groups. (G) KEGG enrichment analysis of differential genes in different risk groups. (H) GSEA analysis of different risk groups



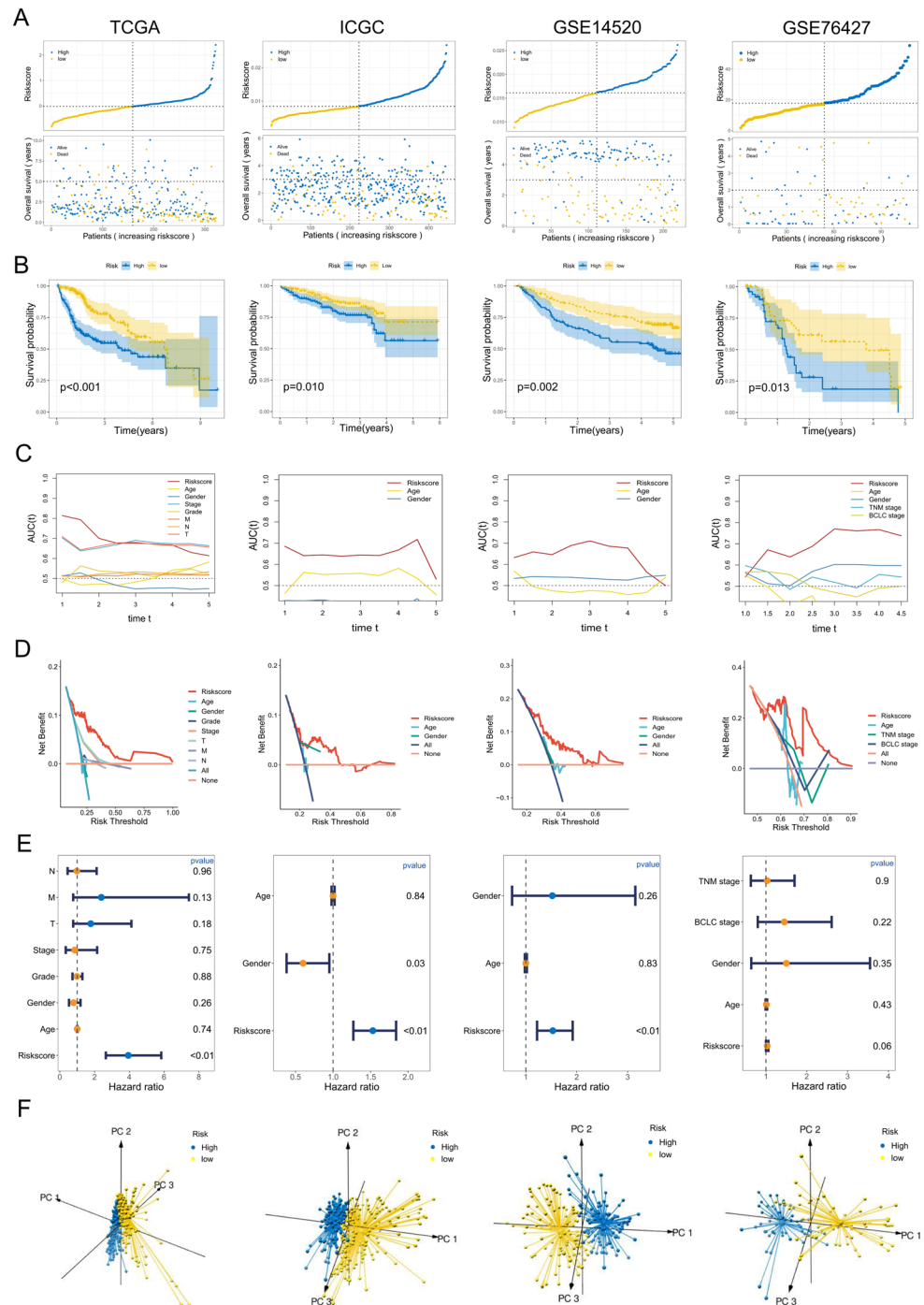
significant upregulation of cell cycle, cellular senescence, and DNA Replication pathways in the high-risk group (Fig. 3H).

Multiple HCC datasets indicate that CS risk score serves as an independent prognostic factor for HCC

In the TCGA and the 3 validation cohorts, as patients' risk score increased, their overall survival time decreased and more patients died (Fig. 4A). Survival analysis showed that

the high-risk group was significantly detrimental to patient prognosis (Fig. 4B). ROC analysis showed that the ability to predict patient prognosis based on the risk score was higher than other clinical characteristics (Fig. 4C). DCA results showed that patients had higher clinical benefits based on the risk score than other clinical characteristics (Fig. 4D). Multivariable Cox regression results showed that the risk score was an independent prognostic factor for patients (Fig. 4E). In addition, PCA analysis showed that the model

Fig. 4 Multiple HCC datasets indicate that CS risk score serves as an independent prognostic factor for HCC. **(A)** Scatter plots of the distribution of survival status and survival time of patients in TCGA, ICGC, GSE14520 and GSE76427 cohorts with increasing the risk score. **(B)** Overall survival analysis of patients in different risk groups in the 4 cohorts. **(C)** Time-dependent ROC curves of the risk score and clinical characteristics of patients in the 4 cohorts. **(D)** The risk score and clinical characteristics of patients in the 4 cohorts DCA curves. **(E)** Multivariate COX regression results of the risk score and clinical characteristics of patients in the 4 cohorts. **(F)** PCA scatter plot based on key genes



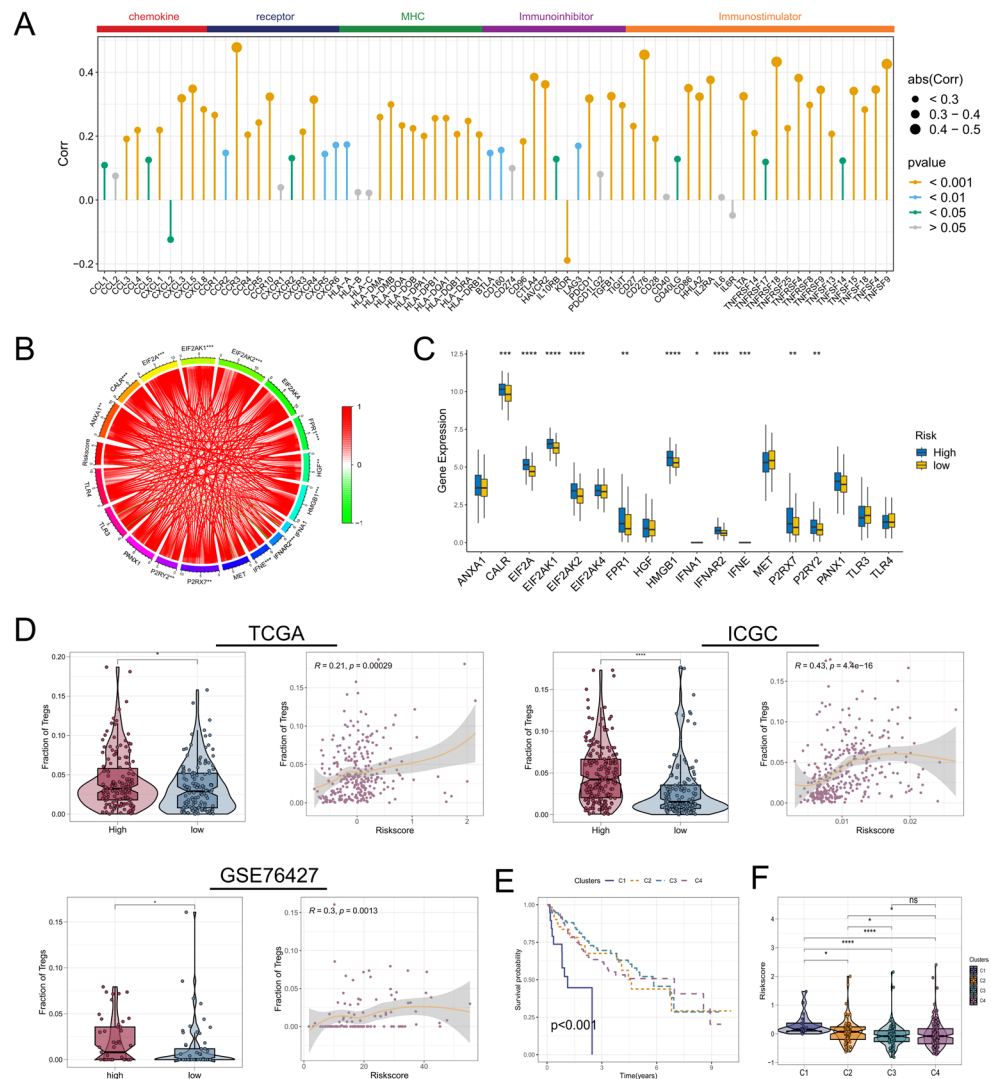
could better differentiate patients into high and low-risk groups (Fig. 4F).

High CS risk score predicts immunosuppressive microenvironment in HCC

There was a significant positive correlation between the risk score and the expression of multiple immunomodulators, such as PDCD1, CD274, and CTLA4 (Fig. 5A, Figure S2A). Immunogenic cell death (ICD)-related genes were also closely associated with tumorigenesis and progression [31], and the expression of multiple ICD genes was significantly upregulated in the high-risk group (Fig. 5B, C). We calculated the correlation between the risk score and immune cell infiltration. The results showed that regulatory T cells (Treg) were significantly positively correlated with the risk score across three algorithms (Figure S3A). In the CIBERSORT algorithm, among the different risk groups of the four datasets, only Treg showed a consistent trend

in immune infiltration proportion (Figure S2B-E), and the infiltration proportion of Treg in the high-risk group was significantly higher than that in the low-risk group, which may partly explain the poorer prognosis of high-risk patients (Fig. 5D). Further analysis revealed that CD4⁺ Th2 cells, M0 macrophages, and M2 macrophages were also significantly positively correlated with the risk score across multiple algorithms (Figure S3B-D), suggesting a more complex immunosuppressive microenvironment in the high-risk group. Subsequently, we classified immune-related subtypes in the TCGA cohort patients and showed that immune-related subtype 1 was detrimental to patient prognosis and that subtype 1 patients had significantly higher risk scores than other subtypes (Fig. 5E), and the relative proportion of subtype 1 was similarly higher in patients in the high-risk group than in those in the low-risk group (Fig. 5F).

Fig. 5 High CS risk score predicts immunosuppressive microenvironment in HCC (A) Correlation of the risk score with the expression of multiple immunomodulators. (B) Chord plot of correlation between the risk score and genes related to immunogenic cell death (positive correlation in red, negative correlation in green). (C) Box plot of differential expression of genes related to immunogenic cell death in different risk groups. (D) Correlation of the risk score with the ratio of Tregs cell infiltration in 3 cohorts. (E) Four immune subtypes in the TCGA cohort. (F) Differential distribution of the risk score of the four immune subtypes

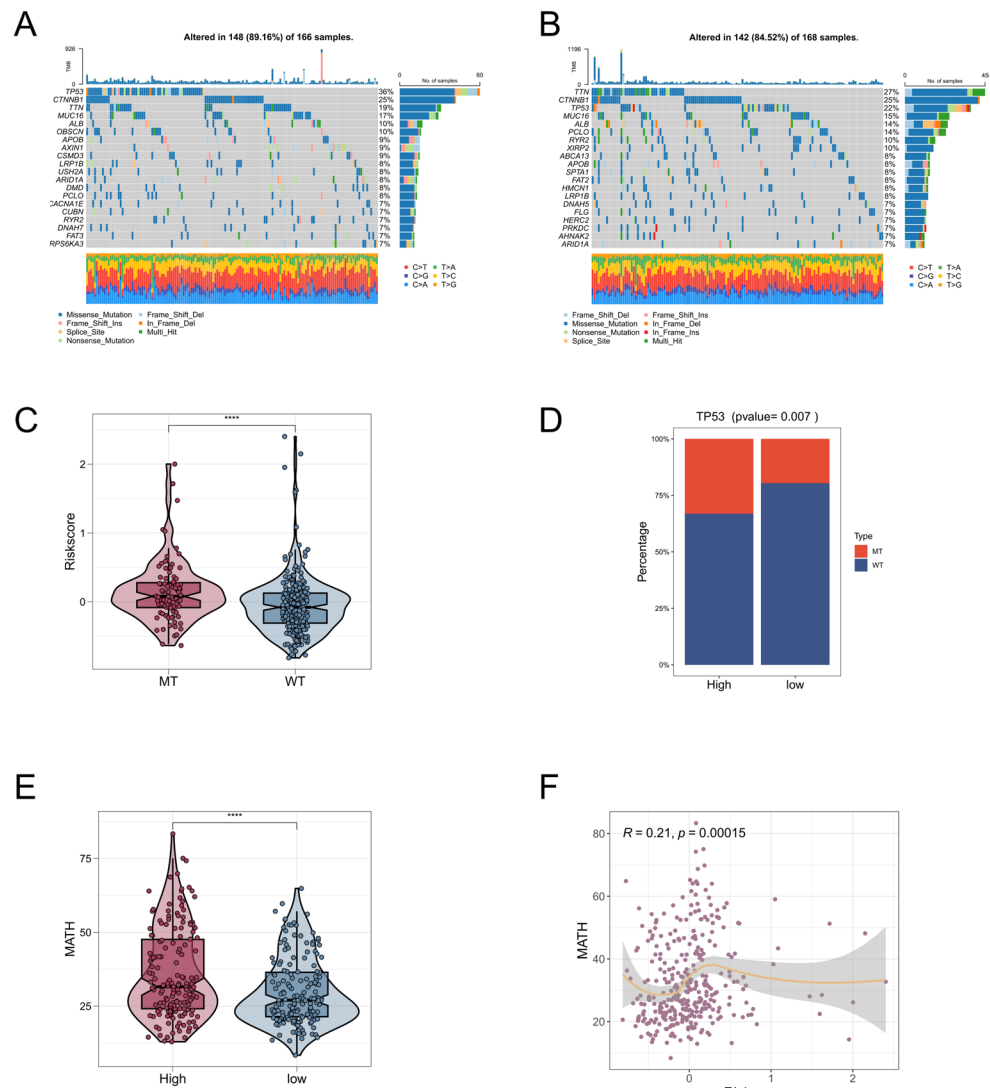


The CS risk pattern has different mutation patterns and heterogeneity in HCC

Gene mutations play an important role in tumorigenesis and progression. The overall mutation frequency was higher in the high-risk group compared to the low-risk group (Fig. 6A-B). Regarding TP53 mutation, the risk score was significantly higher in patients with TP53 mutations, and the proportion of TP53 mutations was significantly higher in the high-risk group than in the low-risk group (Fig. 6C-D). We further analyzed the differences in MATH scores between the risk groups. The MATH score was significantly higher in the high-risk group, and the risk score was positively correlated with the MATH score (Fig. 6E-F). The MATH score, which reflects tumor heterogeneity, may be associated with the poor prognosis of patients in the high-risk group [32].

Fig. 6 The CS risk pattern has different mutation patterns and heterogeneity in HCC.

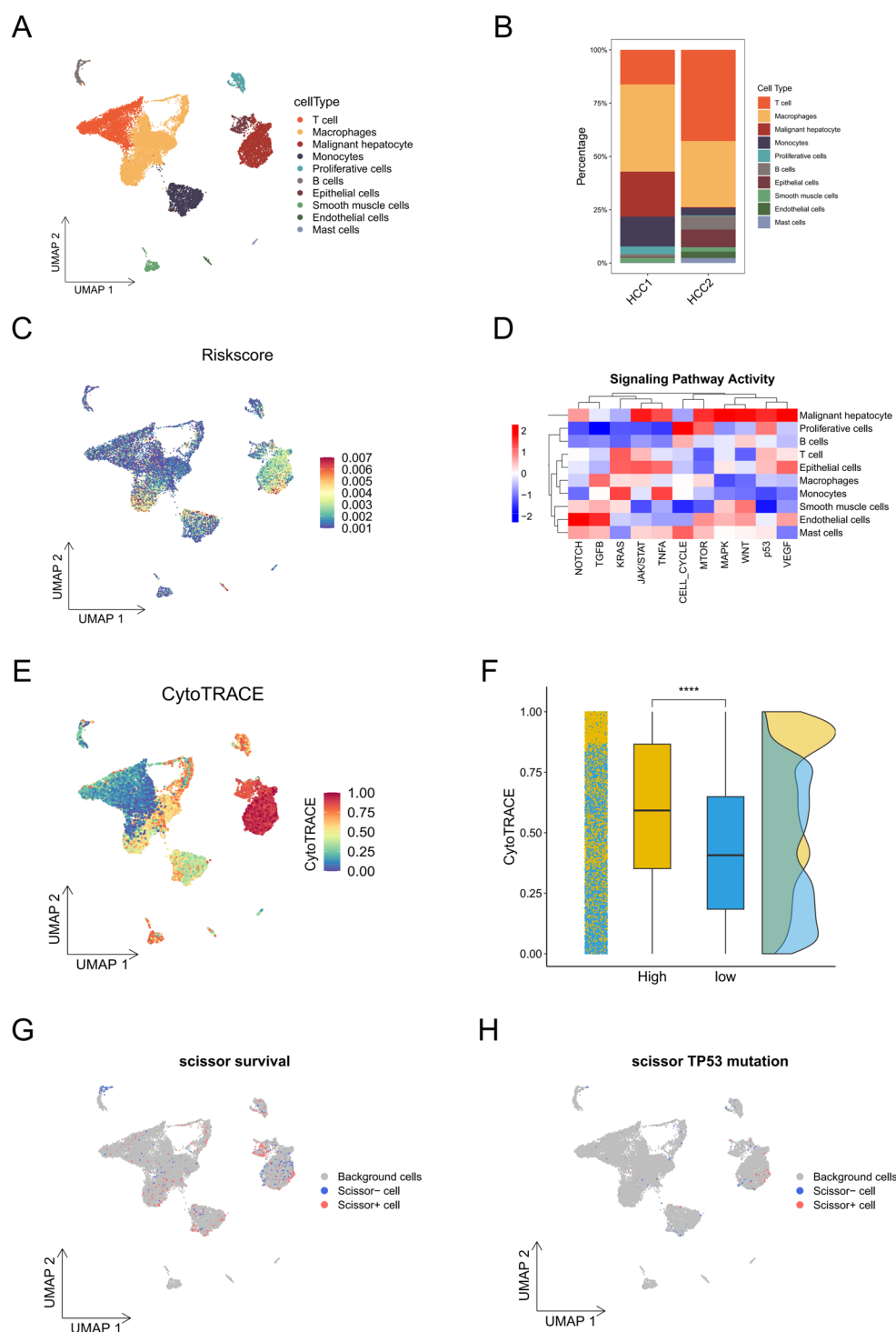
(A) Waterfall plot of the top 20 gene mutations for patients in the high risk group. (B) Waterfall plot of the top 20 gene mutations for patients in the low risk group. (C) Violin plot of the risk score for TP53 mutant and wild type. (D) Distribution of TP53 mutant and wild type in different risk groups. (E) Violin plot of MATH score distribution for different risk groups. (F) Scatter plot of correlation between the risk score and MATH scores



High CS risk score is mainly enriched in HCC malignant cells with single cell sequencing

Subsequently, we further explored the distribution of the risk score among different cells by single-cell data. As shown in Figure S4A and B, all cells were clustered into 12 cell clusters. The 12 cell clusters were annotated into 10 different cell types according to the marker genes of different cell clusters (Figure S4C, Fig. 7A-B). The results of the ROGUE analysis showed that the ROGUE scores of all 10 cell types were around 0.9, indicating the low cell heterogeneity within each cell type derived from our annotation (Figure S4D). And the risk score was mainly expressed in malignant hepatocytes (Fig. 7C). We then analyzed the enrichment activity of each cell with multiple cancer-related pathways, and malignant hepatocytes showed significantly higher enrichment activity in signaling pathways such as VEGF, WNT, and MAPK than other cell types (Fig. 7D). CytoTRACE analysis showed that malignant hepatocytes

Fig. 7 High CS risk score is mainly enriched in HCC malignant cells with single cell sequencing. **(A)** Scatter plot of the descending distribution of UMAP for the 10 cell types identified. **(B)** Proportion of distribution of each cell type in different samples. **(C)** Expression of the risk score in each cell. **(D)** 11 pathway enrichment activities for each cell type. **(E)** Scatter plot of CytoTRACE scores for each cell. **(F)** Box plot of CytoTRACE scores for cells in different risk groups. **(G)** Prognostic status of each cell as inferred by the scissor algorithm. **(H)** TP53 mutation status of each cell as calculated from TCGA cohort data according to the scissor algorithm



exhibited significantly higher stemness scores than other cell types, and CytoTRACE scores were also higher in high-risk group cells than in the low-risk group. This indicates that the risk score is predominantly upregulated in cells with enhanced stemness rather than differentiation (Fig. 7E-F). To further investigate the prognostic impact of malignant hepatocyte infiltration levels in patients, we performed ssGSEA

analysis on TCGA cohort samples using genes significantly upregulated by malignant hepatocytes. The results showed that high infiltration levels of malignant hepatocytes were detrimental to patient prognosis and that their infiltration levels were positively correlated with the risk score (Figure S4E-F). Since the risk score was strongly correlated with poor prognosis and TP53 mutation in patients, we inferred

the survival status and TP53 mutation status of each cell by the scissor algorithm, as shown in Fig. 7G-H, the number of cells associated with poor prognosis as well as TP53 mutation was higher in malignant hepatocytes than in other cell types, which further validated our speculation.

High-risk CS patterns predict poorer drug sensitivity and immunotherapy response

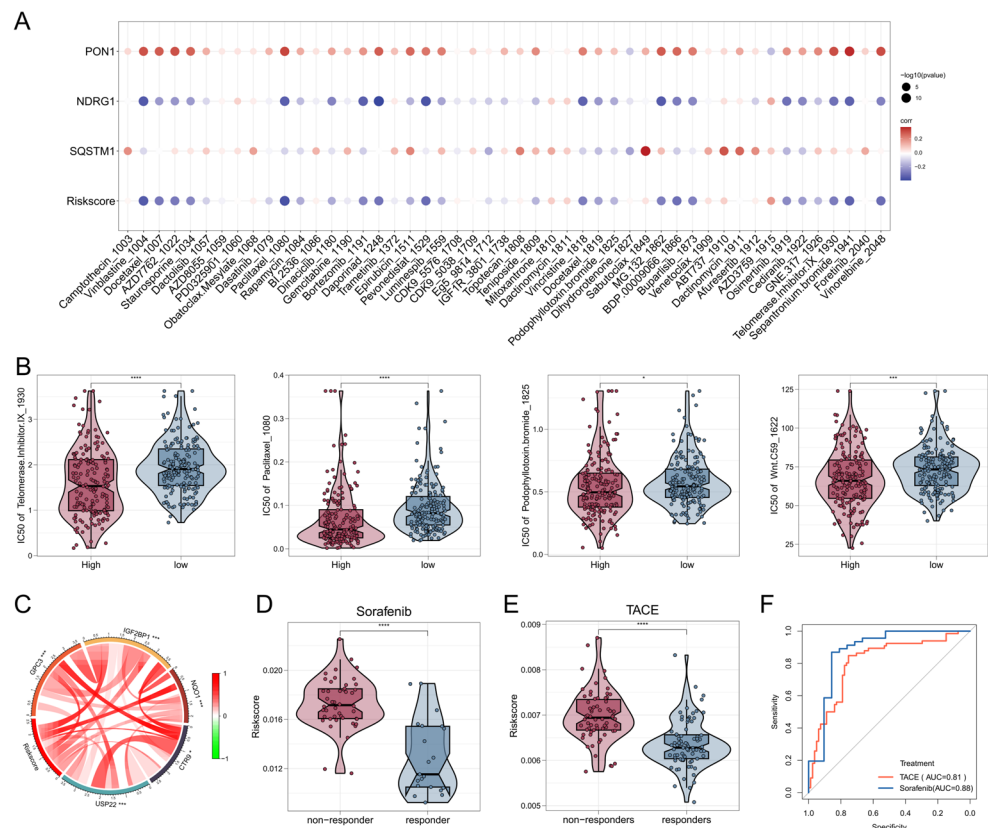
To investigate the relationship between the risk score and sensitivity to multiple drugs, we calculated drug half-maximal inhibitory concentration (IC₅₀) values for each sample in the TCGA cohort. As shown in Fig. 8A and B, the risk score was negatively correlated with the IC₅₀ of multiple drugs, and patients in the high-risk group may have a higher sensitivity to certain drugs, such as Telomerase inhibitor, Paclitaxel, Podophyllotoxin bromide, and Wnt C59. We then collected genes that have been demonstrated by others to be promising for targeting HCC therapy [33–37], and the risk score was positively correlated with the expression of these target genes (Fig. 8C). In addition, Sorafenib and TACE are now commonly used clinical treatments for patients with HCC. In patients treated with Sorafenib and TACE, the risk score was significantly higher in non-responders than in responders, and the risk score was a reliable predictor of treatment response in patients with AUC values above 0.8 (Fig. 8D-F). To further validate the prognostic value of

the risk score in these treatment populations, we analyzed survival outcomes in independent cohorts. In the GSE14520 TACE-treated cohort, high-risk patients exhibited significantly poorer overall survival than low-risk patients (Figure S5C). In contrast, in the TCGA-LIHC sorafenib-treated subgroup, high-risk patients showed a trend toward worse overall survival compared to low-risk patients, although the difference did not reach statistical significance (Figure S5D). Beyond conventional therapies, we also evaluated the predictive value of the CS risk score for immunotherapy response. We analyzed multiple independent external immunotherapy cohorts. In spatial transcriptomic data from HCC patients receiving cabozantinib and nivolumab combination therapy, non-responders exhibited significantly higher risk scores than responders (Figure S5A-B). Similarly, across three additional immunotherapy cohorts (E-MTAB-6270, GSE67501, and PRJEB25780), responders showed significantly lower risk scores compared to non-responders, and low-risk patients were more likely to achieve clinical benefit (Figure S5E-G), suggesting a higher benefit for patients in the low-risk group receiving immunotherapy.

CS prognostic modeling improves the prediction of 1-, 3-, and 5-year survival in HCC patients

Naturally then, we performed a meta-analysis of the prognostic impact of the risk score in the 4 cohorts and found

Fig. 8 High-risk CS patterns hint at poorer drug sensitivity and immunotherapy response. **(A)** Correlation between the risk score and IC₅₀ values of multiple drugs. **(B)** Violin plots of IC₅₀ of telomerase inhibitor, Paclitaxel, Podophyllotoxin bromide and Wnt C59 in different risk groups. **(C)** Chord plot of correlation between the risk score and HCC target genes. **(D)** Distribution of differences in the risk score between responders and non-responders in HCC recipients treated with Sorafenib. **(E)** Distribution of differences in the risk score between responders and non-responders in HCC recipients treated with TACE. **(F)** ROC curve of the risk score predicts response to treatment with Sorafenib and TACE in HCC patients



that high-risk score in all 4 cohorts were factors contributing to poor patient prognosis and that the random effects model was equally significant (Fig. 9A). Subsequently, we constructed a nomogram model for the samples of the TCGA cohort, and high nomogram scores were unfavorable to patient prognosis (Fig. 9B). It had high accuracy in predicting patients' survival at 1, 3, and 5 years (Fig. 9C). the ROC results showed that the AUC values of the nomogram model were higher than other clinical characteristics (Fig. 9D). In Fig. 9E, the benefit rate of nomogram-based patients receiving clinical treatment was higher than other clinical features. In addition, the nomogram model had high predictive ability in the 3 validation cohorts (Fig. 9F-H).

Identical expression of PON1, NDRG1 and SQSTM1 in HCC transcriptome, proteome, immunohistochemistry and spatial transcriptomics

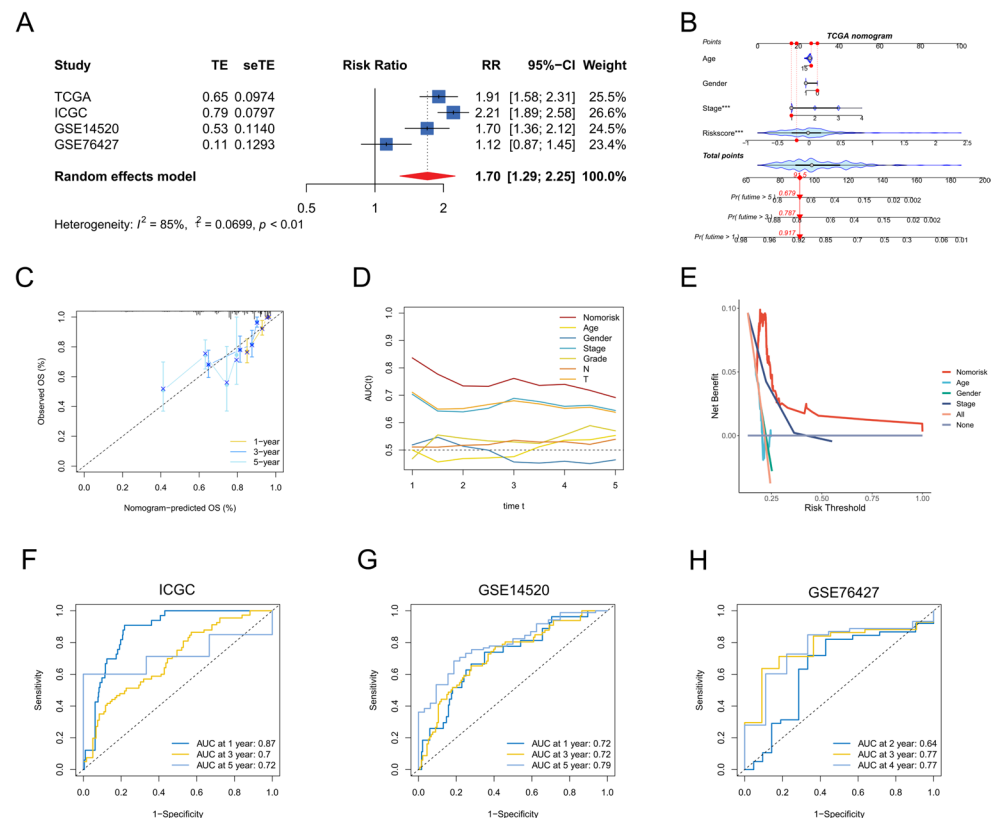
Subsequently, we further investigated the differential expression of the three key genes in HCC and normal tissues. Among them, both mRNA and protein expression levels of the PON1 gene were significantly down-regulated in HCC, while those of NDRG1 and SQSTM1 were significantly up-regulated (Fig. 10A, B). Immunohistochemical staining results from the HPA database further supported this conclusion (Fig. 10C). Spatial transcriptomic analysis further demonstrated that the expression trends of these

genes were consistent with the above findings, and notably, high-risk score regions were predominantly enriched in malignant cell areas (Fig. 11), which was consistent with our single-cell analysis results.

Discussion

Although the availability of various clinical treatments for HCC patients, the low cure and resectability rates are due to the heterogeneity of the tumor and the distinct structural growth patterns observed in HCC [38]. Currently, an increasing number of studies have shown that CS is closely linked to the development and progression of several malignancies, including HCC, and that excessive accumulation of senescent cells in tissues inhibits the growth regulation of other cells and leads to functional decline [39, 40]. Under persistent liver injury, chronic inflammation, and gut microbiota-derived metabolites, senescent hepatocytes progressively accumulate and secrete senescence-associated secretory phenotype (SASP) factors, including IL-1 β , IL-6, CXCL12, TGF- β , CXCL1 and CXCL2. IL-1 β and IL-6 can promote HCC cell proliferation and invasion through activation of the IL-6/STAT3 signaling pathway [41–43]. Furthermore, senescent hepatocytes remodel the tumor immune microenvironment by inducing M2 polarization of macrophages via SASP, contributing to poor prognosis in

Fig. 9 CS prognostic modeling improves prediction of 1-, 3-, and 5-year survival in HCC patients. **(A)** Meta-analysis forest plot of the risk score for the 4 cohorts. **(B)** nomogram model for patients in the TCGA cohort. **(C)** Calibration curves for nomogram scores at 1, 3, and 5 years. **(D)** Time-dependent ROC curves of nomogram scores versus other clinical characteristics. **(E)** DCA analysis of nomogram scores versus other clinical characteristics. **(F–H)** ROC curves of ICGC, GSE14520, and GSE76427 cohorts at 1, 3, and 5 years



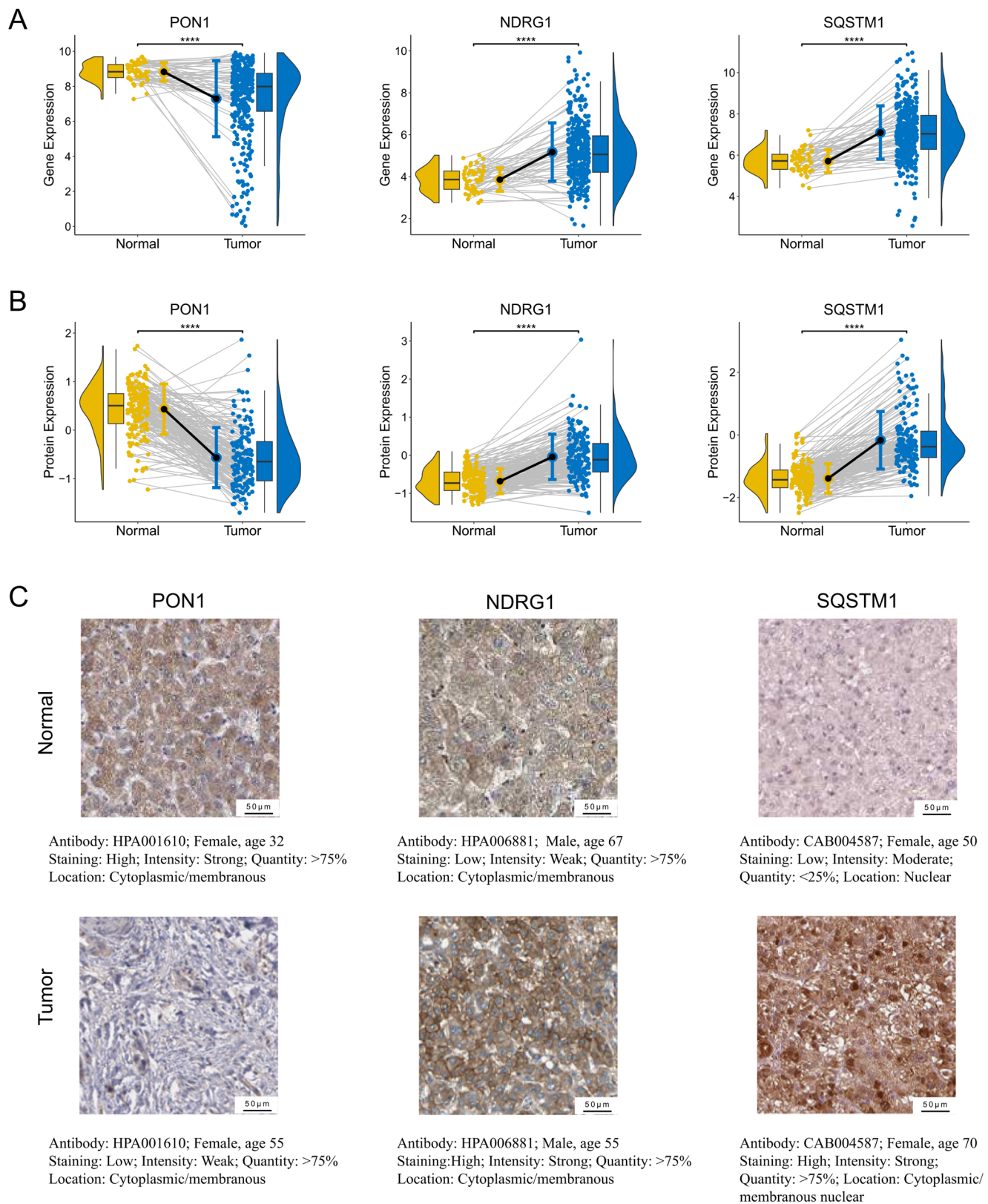
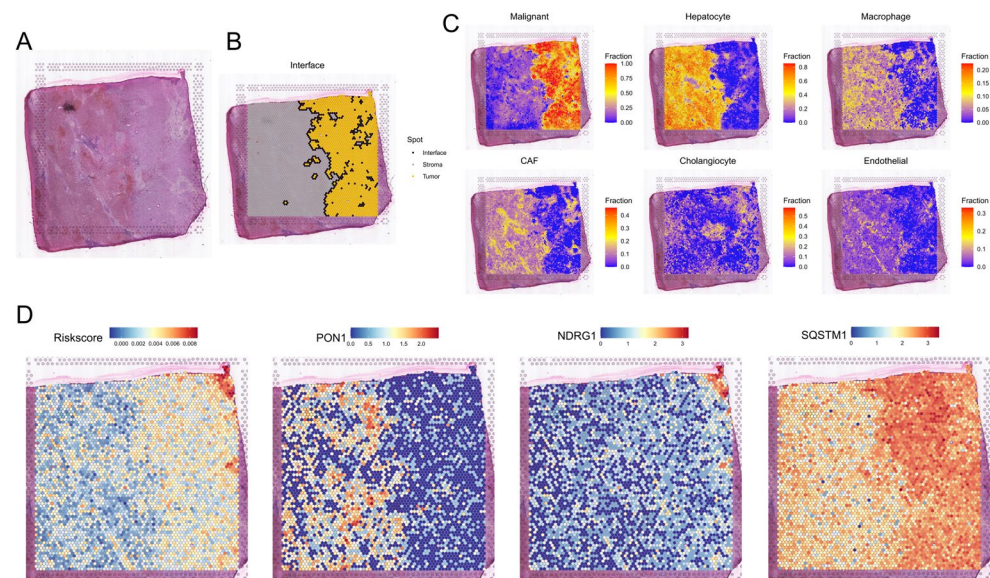


Fig. 10 Consistent expression of PON1, NDRG1 and SQSTM1 in HCC transcriptome, proteome and immunohistochemistry. **(A)** mRNA expression of PON1, NDRG1 and SQSTM1 genes in the TCGA cohort in HCC and normal tissues. **(B)** Protein level expression of PON1,

NDRG1 and SQSTM1 genes in the CPTAC-PDC000198 cohort in HCC and normal tissues. **(C)** Protein level expression of PON1, NDRG1 and SQSTM1 in immunohistochemical staining results in hepatocellular carcinoma and normal tissues

Fig. 11 Consistent expression of PON1, NDRG1 and SQSTM1 in HCC spatial transcriptomics. **(A)** H&E staining image of the tissue section. **(B)** Tumor-stroma interface region. **(C)** Visualization of cell type deconvolution results, including malignant cells, hepatocytes, macrophages, cancer-associated fibroblasts (CAFs), cholangiocytes, and endothelial cells. **(D)** Spatial transcriptomic risk score. **(E)** Expression of PON1, NDRG1 and SQSTM1 in spatial transcriptomics



HCC patients [44]. The miR141-CXCL1-CXCR2 pathway can regulate the migration of Tregs to the tumor microenvironment [45]. Additionally, CXCL12 promotes the migration of Tregs to the tumor microenvironment by binding to the CXCR4 receptor. TGF- β can induce Foxp3 expression in naive T cells, promoting their differentiation into CD4 + CD25+ Tregs; CCL5 can stimulate Tregs proliferation through the RANKL-RANK signaling pathway [46]. The combined action of these pro-inflammatory cytokines leads to chronic inflammation at the tumor site, recruiting large numbers of Tregs and myeloid-derived suppressor cells, which secrete immunosuppressive cytokines such as IL-10, IL-6, and TGF- β , express immune checkpoint ligands such as PD-L1, and inhibit T cell activity through metabolic competition [47]. This chronic inflammatory microenvironment driven by inflammatory and immune regulatory factors closely resembles the typical features of SASP. Therefore, the CS-related risk score may reflect immune remodeling mediated by SASP, ultimately leading to an immunosuppressive state. In this study, the increased Tregs infiltration, elevated immune checkpoint expression, and activation of inflammatory pathways in the high-risk group are consistent with these immunosuppressive characteristics. Thus, we propose that the CS risk score serves as an integrative indicator linking cellular senescence, immune evasion, and poor prognosis in HCC. This SASP-mediated immunosuppressive microenvironment may be a key driver of immune checkpoint inhibitor resistance, and CS-related immune remodeling may predict poor immunotherapy response in HCC patients. Notably, in addition to immune regulation, hepatocyte senescence itself may also trigger mutations, thereby promoting the high invasiveness of HCC [48]. Collectively, these findings suggest that targeting senescent cells or their SASP components may synergize with

immunotherapy to reverse immunosuppression, thereby overcoming resistance and providing a novel therapeutic strategy for HCC.

By exploring CS-related genes differentially expressed in normal and HCC tissues and thereby classifying patients into three subtypes, we screened for CS genes most associated with subtypes with poor prognosis, identified three CS-related genes with independent prognostic value by LASSO and COX regression models, and constructed CS-related risk score from them. Among them, the PON1 gene was expressed down-regulated in HCC tissues and favored patient prognosis, whereas the NDRG1 and SQSTM1 genes were expressed up-regulated in HCC tissues and mediated the poor prognostic characteristics of patients. In the study by CAO et al. [49], it was found that PON1 is a serum enzyme associated with high-density lipoprotein, mainly synthesized and released by the liver, and changes in its serum levels are involved in various diseases such as oxidative stress and inflammation. Similarly, a study by LU et al. demonstrated that PON1 expression was downregulated in HCC tissues, while its high expression was beneficial to patient prognosis [50]. NDRG1, as an N-myc downstream regulatory gene, promotes cancer cell proliferation and metastasis in hepatocellular carcinoma, kidney cancer, and lung cancer [51]. Upregulation of NDRG1 expression in HCC tissues may contribute to poor prognosis by increasing the rate of portal vein invasion [52]. SQSTM1 is a multifunctional scaffolding protein that plays a crucial role in activating multiple signaling pathways, including oxidative stress, selective autophagy, and nutrient metabolism [53]. As an autophagy-related receptor, SQSTM1 binds to polyubiquitinated compounds, thereby promoting the activation of autophagy-related pathways and leading to the degradation of specific substrates [54]. Furthermore, a study by ZHENG

et al. [55] also found that SQSTM1 gene expression was upregulated in HCC tissues and was significantly associated with poor prognosis in patients. It is worth noting that these three genes may work in coordination. PON1 can promote the ubiquitination and degradation of HIF-1 α by recruiting VHL, thereby inhibiting lactate production [50]. In contrast to PON1, SQSTM1/p62 is a strong activator of the Warburg effect, which can inhibit the interaction between VHL and HIF-1 α , enhance the stability of HIF-1 α , and promote lactate production and glucose uptake [56]. Lactate can upregulate the expression of NDRG1 through histone H2B lactylation, enhancing the anti-senescence capacity of HCC cells and ultimately leading to a poor prognosis [57]. In addition to their roles in metabolic-epigenetic regulation, these genes may also be involved in immune modulation. Studies have shown that downregulation of PON1 expression leads to lactate accumulation, thereby promoting the infiltration of Tregs [50]. Moreover, as an antioxidant enzyme, PON1 knockdown disrupts its antioxidant function in cancer cells, enhancing tumor invasiveness via activation of the Akt signaling pathway [58]. Lactate accumulation, which upregulates NDRG1 expression, may also facilitate immune evasion through the promotion of Treg cell differentiation, CD8 + T cell exhaustion, and polarization of M2 macrophages [59]. Overexpression of SQSTM1/p62 directly activates the NF- κ B pathway and increases production of pro-inflammatory cytokines such as IL-1 β , IL-6, and TNF- α , thereby affecting immune cell recruitment [60]. In summary, PON1, SQSTM1, and NDRG1 collectively influence the prognosis of HCC by synergistically regulating lactate metabolism, epigenetic modifications, and immune-related pathways. This mechanistic framework provides a rational basis for understanding the biological underpinnings of the CS risk score, although the specific regulatory network still requires further validation through *in vitro* and *in vivo* experiments.

TP53 is a key regulatory factor in DNA damage response, cell cycle arrest, apoptosis, and cell senescence. Under normal physiological conditions, p53 can promote cell senescence through pathways such as p53/p21-mediated cell cycle arrest [61]. However, TP53 mutations can disrupt this typical senescence checkpoint, allowing tumor cells to evade normal senescence control and accumulate DNA damage, genomic instability, and events related to malignant progression [62]. Therefore, the enrichment of TP53 mutations in the high CS risk group may indicate that this group represents an imbalanced or unadaptive state of aging rather than a protective physiological aging state. In hepatocellular carcinoma (HCC), TP53 mutations are usually associated with chromosomal instability and changes in the tumor immune microenvironment [63]. Thus, the association between high CS risk and TP53 mutations that we observed may be caused

by defective p53 signal transduction leading to senescence escape, persistent cellular stress, inflammatory remodeling, and changes in the immune microenvironment. However, we acknowledge that this remains a hypothesis based on integrative bioinformatic analysis, and further mechanistic studies are needed for verification.

Our research revealed that patients with a high-risk score exhibited a markedly unfavorable prognosis, which may be intricately linked to their heightened pro-tumor immunity (including greater infiltration of Tregs), and increased production of gene mutations. A study by XIE et al. [64] discovered that the upregulation of Tregs cell infiltration in HCC tissues was connected to the emergence of immune checkpoint inhibitor resistance. Furthermore, in the realm of antitumor therapy, sorafenib and TACE represent the primary drugs and approaches commonly employed in clinical practice for treating patients with HCC [65, 66]. Patients who exhibited an unresponsive reaction to sorafenib and TACE treatment had a significantly higher risk score, suggesting that the risk score may be associated with clinical response to sorafenib and TACE in HCC patients. The application of immunotherapy in oncology treatment is currently progressing rapidly, with preliminary outcomes demonstrating benefits for numerous patients afflicted with solid or hematologic tumors [67, 68]. Nonetheless, there remain certain patients who exhibit unfavorable outcomes with regard to immunotherapy, potentially owing to mechanisms such as their low mutational load and limited expression levels of immunomodulators [69, 70]. This study observed upregulated expression of multiple immune checkpoint genes in patients within the high-risk group, implying that patients in this group may potentially exhibit a greater benefit rate for immunotherapy. To further validate this, we analyzed real-world immunotherapy datasets, and consistently found that responders had significantly lower risk scores than non-responders, with low-risk patients showing greater benefit from immunotherapy. Furthermore, notable variations in the expression of immunogenic cell death genes were evident among patients belonging to distinct risk groups, underscoring the potential utility of risk score-based investigations as a guideline for the clinical treatment of HCC patients.

Despite the significant findings presented in our study, several limitations warrant further investigation. Specifically, our study lacks experimental validation of the expression status of the three CS-related genes in both HCC and normal tissues, and there is a need for relevant cellular and animal experiments to elucidate the underlying molecular regulatory mechanisms. Furthermore, several drugs predicted in this study have not yet received FDA or NMPA approval for the clinical treatment of HCC, and the relevant conclusions require further validation in clinical trials. In addition, our spatial transcriptomics analysis was based

on a single tissue section, warranting validation in larger, clinically annotated cohorts. Moreover, the findings on immunotherapy response prediction require further confirmation in larger, dedicated HCC immunotherapy cohorts. Additionally, it remains unclear whether the CS risk score directly regulates the immune microenvironment or merely correlates with it. The observed correlations with immune infiltration, immune checkpoint expression, and mutational features may reflect intrinsic molecular changes such as chronic inflammation and hypoxia, common upstream drivers, or differences in cellular composition across tumor samples. Therefore, the CS risk score should be regarded as a comprehensive indicator reflecting the interaction between HCC biology and the immune microenvironment, rather than a proven causal regulator. Further mechanistic studies including single-cell RNA sequencing, spatial transcriptomics, and perturbation experiments targeting CS-related genes are needed to determine whether and how the CS risk score is directly associated with immune reprogramming in HCC.

Additionally, as a retrospective bioinformatic study, our findings are derived from public datasets, which may introduce potential biases such as patient selection, platform heterogeneity, and variability in clinical annotation across cohorts. Although multi-cohort validation mitigated these concerns, prospective studies are needed to confirm the clinical value of the CS-related risk score.

Conclusions

The CS-related risk score may play an important role in the development and progression of HCC, providing a reference for determining the clinical treatment and prognosis of HCC patients.

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Author contributions Jin Lu and Junjie Hu conceived the research idea and participated in the overall revision and polishing of the manuscript. Shaoyang Lu and Junjie Ma were responsible for data analysis and manuscript writing. Xiaodan Wang prepared the figures and contributed to the overall revision. Lei Zhang completed the background and portions of the discussion. All authors read and approved the final manuscript.

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Data availability The Bluk, Single-cell transcriptomic and Spatial

Transcriptomics data analyzed in this study can be found at TCGA: <https://portal.gdc.cancer.gov/>. GEO: <https://www.ncbi.nlm.nih.gov/gene/>. ICGC: <https://docs.icgc-argo.org/docs/data-access/icgc-25k-data>. TIGER: <http://tiger.canceromics.org/>. CROST: <https://ngdc.cnbc.ac.cn/crost/home>. The protein expression data analyzed in this study can be found at CPTAC: <https://pdc.cancer.gov/pdc/browse>. The CS-related genes can be found at HAGR: <https://genomics.senescence.info/>.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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